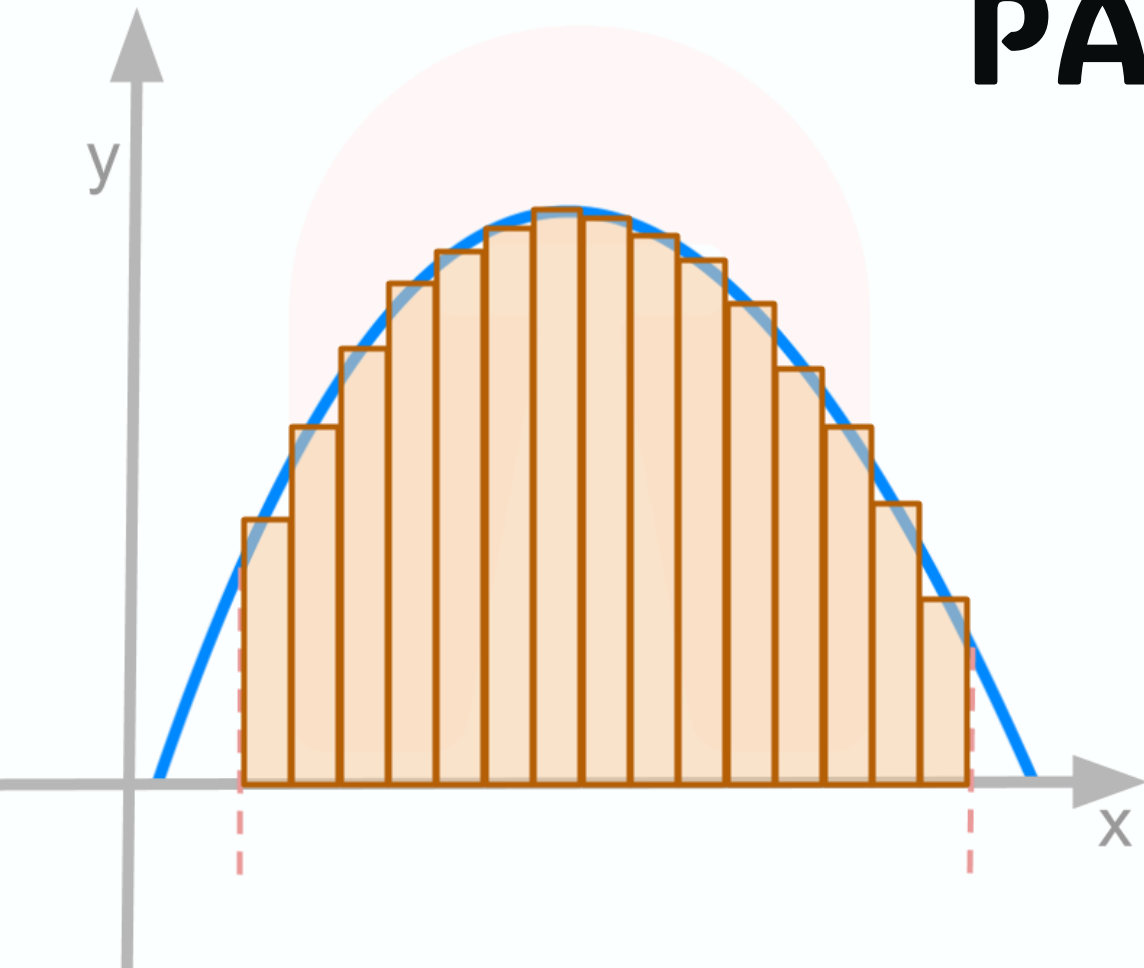


CALCULUS – INTEGRATION

Chapter 15

PAGE 371



$$\int_a^b f(x) dx$$

15.1 INTEGRATION – DIFFERENTIATION REVERSED

KEYWORDS (P.371)

Integration

积分

Indefinite integral

不定积分

Definite integral

定积分

15.6 INTEGRATION OF FUNCTIONS IN THE FORM $\frac{1}{x}$ AND $\frac{1}{ax+b}$

CLASS DISCUSSION (P.384)

Raju and Sara are asked to find $\int \frac{1}{2(3x-1)} dx$

Raju writes: $\int \frac{1}{2(3x-1)} dx = \int \frac{1}{6x-2} dx = \frac{1}{6} \ln(6x-2) + c$

Sara writes: $\int \frac{1}{2(3x-1)} dx = \frac{1}{2} \int \frac{1}{3x-1} dx = \left(\frac{1}{2}\right) \left(\frac{1}{3}\right) \ln(3x-1) + c$

Decide who is correct and discuss the reasons for your decision with your classmates.

$$= \frac{1}{6} \ln(3x-1) + c$$



15.9 FURTHER DEFINITE INTEGRATION

EXAMPLE 18 (W.E.18, P.394)



a) Given that $y = x \cos 2x$, find $\frac{dy}{dx}$

b) Hence evaluate $\int_0^{\frac{\pi}{6}} 2x \sin(2x) dx$

15.9 FURTHER DEFINITE INTEGRATION

EXAMPLE 18 (W.E.18, P.394)



a) Given that $y = x \cos 2x$

a) $\frac{dy}{dx} = \cos(2x) - 2x \sin(2x)$

b) $\frac{3\sqrt{3}-\pi}{12}$

b) Hence evaluate $\int_0^{\frac{\pi}{6}} 2x \sin(2x) dx$

15.9 FURTHER DEFINITE INTEGRATION (EX.2)

CHECK YOUR UNDERSTANDING

Answer on the
next slide...



2) a) Given that $y = (1 + e^x)^4$, find $\frac{dy}{dx}$

b) Hence evaluate $\int_1^2 e^x (1 + e^x)^3 dx$

1.	1189
2.	1190
3.	1191
4.	1188
5.	None of the above

15.9 FURTHER DEFINITE INTEGRATION (EX.2)

CHECK YOUR UNDERSTANDING



2) a) Given that $y = (1 + e^x)^4$, find $\frac{dy}{dx}$

b) Hence evaluate $\int_1^2 e^x (1 + e^x)^3 dx$

1.	1189
2.	1190
3.	1191
4.	1188
5.	None of the above